

NETWORK DEVICES AND SELECTIONS

1. Research two actual network switches available in India (for example, from Cisco, TP-Link, or D-Link) and list their key specifications such as number of ports, speed, and price in a table.

-> **Two actual network switches available in India**

Brand / Model	Ports	Speed	Main Features	Approx. India Price
TP-Link TL-SG1024D	24 RJ45	10/100/1000 Mbps	Unmanaged, plug-and-play, rack-mountable, 48 Gbps switching capacity	₹6,799 approx.
TP-Link TL-SG3452	48 RJ45 + 4 SFP	10/100/1000 Mbps	L2 managed, VLAN, QoS, static routing, Omada management	₹23,199–₹24,485 approx.

2. Create a device selection matrix comparing at least one switch, one router, and one firewall that could be used for a small startup office (20-30 employees). Include columns for brand/model, key specs, price, and your rating for each device.

Hint: Use online sources like Amazon India, Flipkart, or manufacturer websites for real-world pricing and specs.

-> **Device selection matrix — small startup office**

For a **20–30 employee startup**, I would use a Gigabit managed switch, business router, and dedicated next-generation firewall.

Device	Brand / Model	Key specifications	Approx. India price	Suitability rating*
Switch	TP-Link TL-SG3452	48× Gigabit RJ45, 4× Gigabit SFP, VLAN, QoS, static routing, Omada management	₹23,199–₹24,485	9/10
Router	TP-Link ER605	5 Gigabit ports, up to 3 WAN connections, load	₹4,703–₹6,076	8.5/10

Device	Brand / Model	Key specifications	Approx. India price	Suitability rating*
		balancing, VPN, VLAN, firewall policies		
Firewall	Fortinet FortiGate 30G	NGFW, IPS, application control, web filtering, VPN, SD-WAN; up to 500 Mbps threat protection	~₹49,599	9/10

3. Imagine you are setting up the network for a new gaming café with 25 computers and public Wi-Fi. Select one switch, one router, and one firewall model for this scenario and justify your choices in 2-3 lines each.

-> **Gaming café — 25 computers + public Wi-Fi**

Selected switch: TP-Link TL-SG3452

The **48-port Gigabit managed switch** gives enough ports for the 25 gaming PCs while leaving substantial room for access points, CCTV, POS systems and future computers. VLAN support is particularly useful for separating gaming PCs, staff/POS devices and public Wi-Fi traffic.

Selected router: TP-Link ER605

The ER605 provides Gigabit connectivity and supports multiple WAN connections, so the café can use two Internet connections for load balancing or backup. Its VLAN, VPN and security features are useful for separating the café's different network segments.

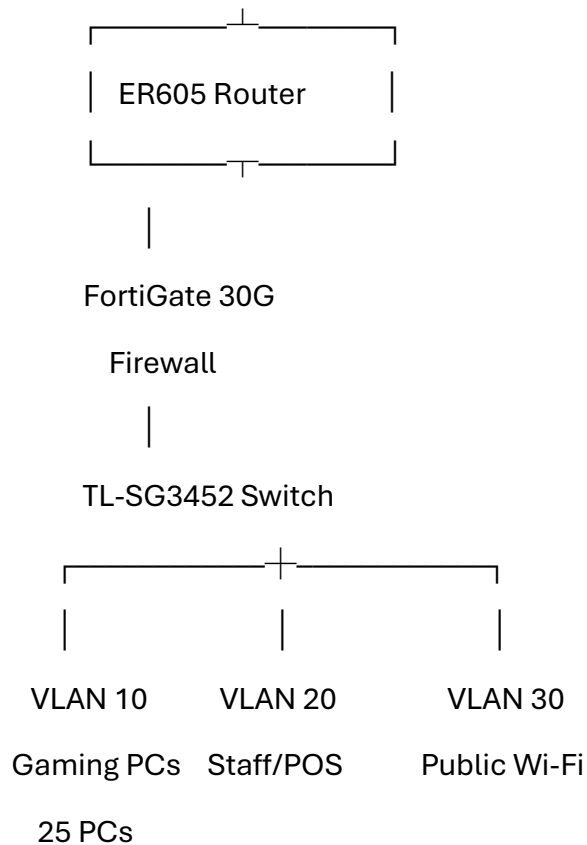
Selected firewall: Fortinet FortiGate 30G

The FortiGate 30G adds dedicated next-generation firewall protection, IPS, application control, web filtering and VPN capabilities. This is useful for a café offering public Wi-Fi because customer traffic can be isolated from the gaming and business networks.

Suggested logical network

INTERNET

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This segmentation helps prevent public Wi-Fi users from directly accessing the gaming PCs or internal business devices.

4. Prepare a Bill of Materials (BOM) listing the selected network devices (switch, router, firewall) for your gaming café scenario, including quantity, model, unit price, total price, and a short justification for each item.

-> Bill of Materials (BOM)

Using representative current Indian retail prices:

Qty.	Device	Model	Unit Price	Total	Justification
1	Managed Switch	TP-Link TL-SG3452	₹23,199	₹23,199	48 Gigabit ports provide capacity for 25 PCs plus Wi-Fi APs and future expansion.
1	Business Router	TP-Link ER605	₹4,703	₹4,703	Supports Gigabit WAN/LAN, multiple WAN connections, VPN

Qty.	Device	Model	Unit Price	Total	Justification
					and VLAN features.
1	NGFW Firewall	Fortinet FortiGate 30G	₹49,599	₹49,599	Provides dedicated security, IPS, web filtering, application control and SD-WAN.
		Estimated total		₹77,501	

5. Perform a cost-benefit analysis for your chosen devices: explain in 3-4 lines why you selected these models over cheaper or more expensive alternatives, considering both features and budget.

->Cost-benefit analysis

The **TL-SG3452** costs more than a basic 24-port unmanaged switch, but the additional ports and VLAN/management capabilities are valuable in a gaming café with public Wi-Fi. A cheaper TL-SG1024D is simpler and less expensive, but it lacks the managed-network capabilities needed for proper segmentation.

The **ER605** provides business networking features at a relatively low cost, including multiple WAN support and VPN/VLAN capabilities.

The **FortiGate 30G** is the largest part of the budget, but it provides security capabilities beyond a basic router firewall, including IPS, application control and web filtering.

Final estimated hardware budget: ₹77,501

For a 25-PC gaming café, the main benefit of this combination is that it balances **Gigabit wired performance, network segmentation, Internet redundancy and dedicated security** without moving to much more expensive enterprise-class hardware.